



Physical and numerical investigation on nonlinear mechanical properties of deep-buried rock tunnel excavation unloading under complicated ground stresses

Wuqiang Cai^a, Hehua Zhu^{a,b,*}, Wenhao Liang^{a,c,*}, BaThao Vu^a, Wei Wu^a

^a Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China

^b State Key Laboratory for Disaster Reduction in Civil Engineering, Tongji University, Shanghai 200092, China

^c China Railway Construction Co, Ltd, Beijing 100855, China

ARTICLE INFO

Keywords:

Complicated geostresses
Deep tunnels
Physical model tests
GZZ-based numerical calculations
Nonlinear mechanical properties

ABSTRACT

The nonlinear stress and deformation response of tunnel excavation unloading under high and complicated geostresses are important features of deep rock tunnel engineering, and such conditions differ significantly from those of shallow tunnels. Additionally, several studies on deep tunnels have focused on mechanical behaviors of rock tunnels at certain depths, ignoring nonlinear responses of rock tunnels at various depths. In this study, physical model tests and three-dimensional (3D) numerical calculations concerning complicated ground stresses were conducted to investigate the nonlinear mechanical and strength properties of tunnels excavated in rock formations with varying buried depths. The generalized Zhang–Zhu criterion (GZZ) was found to appropriately describe the nonlinear laws of rock strength (yield) with stress levels on the π plane, and the correctness and reliability of the GZZ strength criterion were subsequently verified by comparing the deformation monitoring data of physical model testing with different strength criteria-based numerical results. These results were then used to examine the stress distributions, deformations, and failure characteristics of physical model tunnels under complicated stress conditions at various buried depths. The nonlinearity of stress and deformation of the model tunnels are directly related to stress conditions, especially to the horizontal tectonic stress. This indicates that the traditional linear design approaches and practical experiences related to shallow tunnels dominated by gravity stress are not applicable to deep-buried tunnels under high and complicated ground stress conditions.

1. Introduction

Deep rock engineering involves complicated geomechanical environments with high ground stress, high karst water pressure, high temperature, strong disturbance, and strong rheology, which makes the mechanical and strength behaviors of deep rock masses significantly different from those of shallow ones (Li et al., 2014; Yan et al., 2015; Qian and Zhou, 2018; Wagner, 2019; Guo et al., 2020; Zhu et al., 2023). The massive and long-term in situ stress measurements indicate that the ground stresses in deep rock engineering present evident asymmetrical 3D stress conditions ($\sigma_1 > \sigma_2 > \sigma_3$) (Haimson, 1978; Kang et al., 2010; Chen et al., 2017; Wang et al., 2020) and that the orientation and magnitude of the in situ geostresses dramatically change with change in stress concentrations and distributions during excavation unloading, which aggravates the complexity of the stresses around the tunnel.

Compared to shallow rock engineering, the effect of stress redistribution caused by excavation on the stability of deep rock masses is more prominent (Hudson, 1992; Qi et al., 2004). A shallow-buried unloading rock mass is mainly in a linear elastic state, which is adequately explained by the traditional linear theory. In contrast, rock bursts and large deformation disasters are often encountered during the construction of a deep rock tunnel (Meng et al., 2013; Ma et al., 2015; Qian and Zhou, 2018; Chen et al., 2019; Feng et al., 2019; Gong et al., 2019; Zhu et al., 2019; Li et al., 2020; Wu et al., 2023) owing to the nonlinear mechanical response of unloading rock mass under the high and complicated geostresses acting in a deep-buried environment (Singh et al., 1989; Hoek and Brown, 1997; Xie et al., 2019). He et al. (2005) and Xie et al. (2019) investigated the engineering strength and mechanical differences of deep- and shallow-buried rock masses, and confirmed that the mechanical system faced by the former is a nonlinear system. Thus, most linear theoretical studies on shallow tunnels cannot

* Corresponding authors at: Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China.

E-mail addresses: zhuhehua@tongji.edu.cn (H. Zhu), liangwh65130@yahoo.cn (W. Liang).

<https://doi.org/10.1016/j.tust.2023.105197>

Received 11 April 2021; Received in revised form 2 January 2023; Accepted 25 April 2023

Available online 5 May 2023

0886-7798/© 2023 Elsevier Ltd. All rights reserved.

Nomenclature

List of Symbols

k	Lateral pressure coefficients
τ, σ_n	shear and normal stresses
$\sigma_1, \sigma_2, \sigma_3$	Maximum, intermediate principal stress and minimum principal stress
I_1, I_2, I_3	First, second and third stress invariant
J_2, J_3	Second and third deviatoric stress invariant
θ_σ	Lode angle
σ_c	Unconfined compressive strength (UCS)
m_i	Material constant for intact rock
m_b, s, a	Strength parameters for rock masses
GSI	Geological strength indicator
D	Disturbance coefficient
τ_{oct}	Octahedral shear stress
$\sigma_{m,2}$	Effective mean stress, $\sigma_{m,2} = (\sigma_1 + \sigma_3)/2$
c, φ	Rock cohesion and internal friction angle
ρ, E, ν	Density, Elastic modulus and Poisson's ratio
EDZ	Excavation damage zone
R^2	Square of correlation coefficient

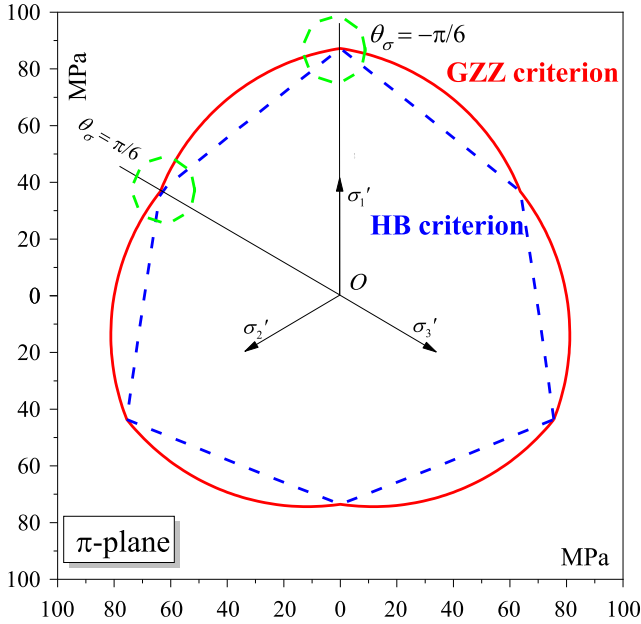


Fig. 1. HB and GZZ strength criteria plotted on the π plane (Cai et al., 2021).

precisely explain the effects of the complicated and high geological stress conditions of deep rock tunnels with respect to their mechanical and engineering characteristics. It is thus clear that the nonlinearities of stress and deformation subjected to complex unloading paths under complicated and high geostresses of deep tunnels cannot be neglected, which constitutes the foundation of stability analysis and operation of a deep tunnel.

However, a critical challenge is to ascertain the stress and deformation behaviors of rock tunnels at different depths under in situ geological conditions. In this regard, physical model tests and numerical simulations that simulate the unloading stress paths are important and effective approaches to recognizing the unloading deformation and failure mechanism of rock tunnels under various buried depths. The strength (yield) criterion is the foundation for judging the elastic-plastic

state of stress and deformation of the rock mass, and the determination of rock mass strength under complicated stress conditions is the research focus in case of a deep rock mass (Hoek and Brown, 1997; He and Qian, 2010; Wang et al., 2010). Two-dimensional (2D) strength criteria (e.g., Hoek–Brown (HB) and Mohr–Coulomb (MC)) are commonly used in the stability analysis of rock mass engineering (Hoek and Brown, 2019). However, deep unloading rock masses exhibit complicated 3D stress conditions, and the design theories of shallow ones, which are dominated by self-weight stress and ignore the influence of intermediate principle stress (σ_2), are not suitable for deep tunnels as it is indisputable that σ_2 has an important influence on the stability of rock and rock mass (Mogi, 1967; Al-Ajmi and Zimmerman, 2005; Pan et al., 2012; Ma and Haimson, 2016; Wang et al., 2020; Zhu et al., 2023). Moreover, there is no evident difference in numerical results obtained using the complex stress conditions of deep tunnels, and the self-weight stress conditions of shallow tunnels when the 2D theory is used. This raises questions regarding its reliability and accuracy. The effect of σ_2 becomes increasingly significant with the increase of burial depth; this shows significant calculation differences based on 2D and 3D strength criteria (Cai et al., 2022a, b, c; Cai et al., 2023; Zhu et al., 2023). Therefore, a reliable 3D strength criterion that is suitable for complicated stress conditions should be adopted for refined analysis and design of deep rock tunnels.

Previously, many researchers have focused on specific engineering cases while investigating the mechanical and strength characteristics of rock tunnels; thus, the obtained stresses and deformations only reflect rock mass behaviors at certain depths. However, few studies have investigated the nonlinear mechanical and strength characteristics of rock tunnels under various buried depths, including the distribution of the plastic zone, the transfer of the principal stress axis, and the nonlinear deformation of surrounding rock governed by tunnel excavation under complex stress conditions and various buried depths. Further study of variations with depth will be beneficial to the stability design and control of deep tunnels. Thus, in this study, the reliability of the generalized Zhang–Zhu (GZZ) strength criterion were investigated and verified by comparing the calculation results of different strength criteria with the physical model test. Model tests and the GZZ-based numerical calculation of model tunnels with different lateral pressure coefficients and different buried depths were then conducted to examine the progressive deformation and failure laws of open-hole tunnels as well as to examine the distribution of the plastic zone, the transfer law of the principal stress, and the nonlinear deformation and failure characteristics of deep tunnel excavation.

2. Rock (Rock mass) strength criteria

In this study, the MC, Drucker–Prager (DP), HB, and 3D nonlinear Hoek–Brown (GZZ) criteria were compared and used to analyze the stress and the deformation of the physical model tunnels under high and complicated ground stresses. These criteria are briefly described in this section.

2.1. MC and DP strength criteria

The explicit MC criterion expressed by principal stress is

$$\sigma_1 = \frac{2c \cos \varphi}{1 - \sin \varphi} + \frac{1 + \sin \varphi}{1 - \sin \varphi} \sigma_3 \quad (1)$$

It is often necessary to express the strength criteria and constitutive models in a unified manner in the form of invariants for the convenience of elastic–plastic finite element calculation. And the MC criterion in the form of invariants is

$$-\frac{1}{3} I_1 \sin \varphi + \left(\cos \theta_\sigma - \frac{1}{\sqrt{3}} \sin \varphi \sin \theta_\sigma \right) \sqrt{J_2} - c \cos \varphi = 0 \quad (2)$$

The strength parameters of MC (cohesion c and internal friction

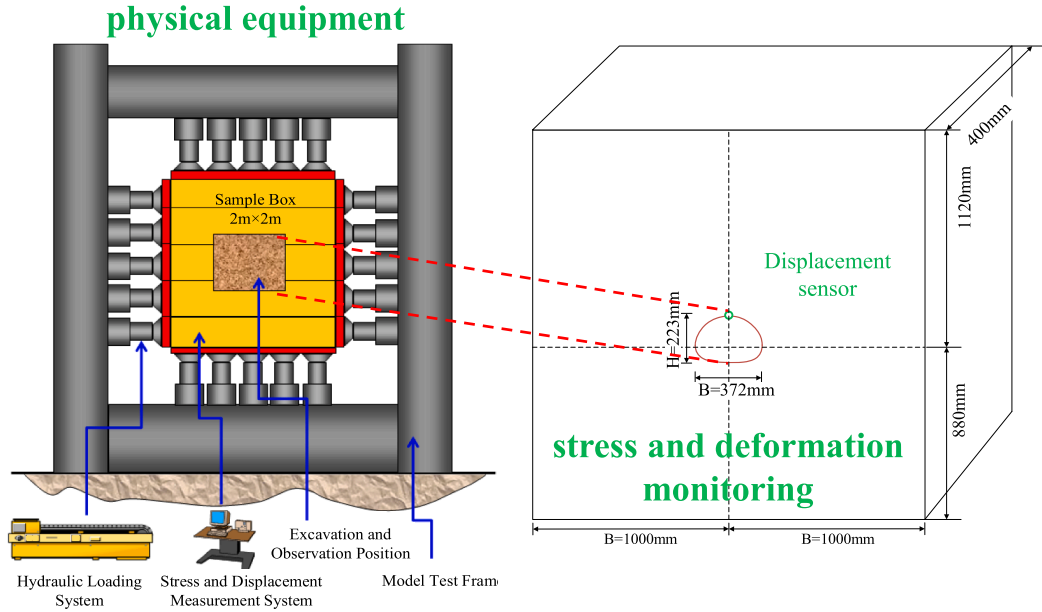


Fig. 2. Model test settings, modified from Vu et al. (Vu, 2014; Vu et al., 2016).

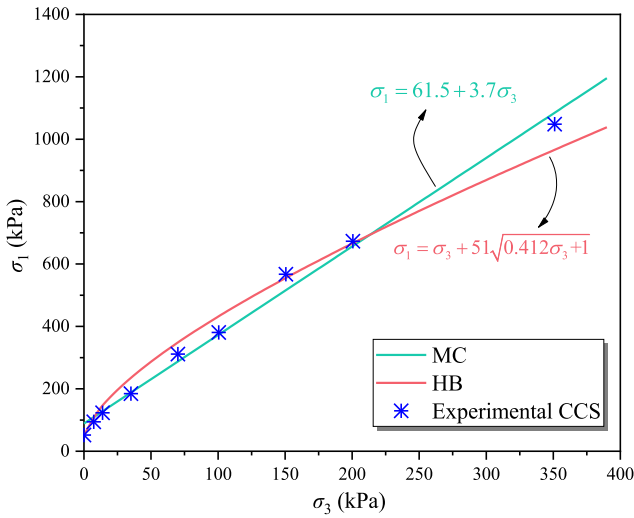


Fig. 3. Fittings of MC and HB strength criteria to triaxial compressive strength (TCS) data of model rock.

Table 1
HB parameters (m , s , a) based on the full-fitting and correlate-fitting.

Different approaches	σ_c MPa	m [13–21]	s [0–1]	a [0.5–1]	GSI [0–100]	R^2
Full-fitting (Approach #1)	0.051	5.47	1.14	0.716		0.998
Correlate-fitting (Approach #2)	0.051	21.4	1.05	0.50	100.5	0.992

Table 2
Physical and mechanical parameters of the model rock.

Parameters of Model rock	E GPa	ρ kg/m ³	c MPa	φ	ν	σ_c MPa	m_b	s	a	GSI	m_i	D
Value	0.09	2100	0.016	35	0.32	0.051	21	1	0.5	100	21	0

angle φ) obtained from indoor test are significant different from those of in situ practical rock mass because of the size effect of rock and rock mass. The rock mass parameters provided by geological exploration data are often fuzzy and semi-qualitative, which may cause huge divergences between numeral results and field practice. Furthermore, the linear MC criterion is not suitable for describing the nonlinear relation between strength of engineering rock mass and confining pressure.

The DP criterion represented by stress invariant is

$$\sqrt{J_2} - aI_1 - k = 0 \quad (3)$$

According to the position relation with MC, parameters $a = \frac{2\sin\varphi}{\sqrt{3}(3-\sin\varphi)}$, $k = \frac{6c\cos\varphi}{\sqrt{3}(3-\sin\varphi)}$ are for the outer circumcircle DP, while $a = \frac{2\sin\varphi}{\sqrt{3}(3+\sin\varphi)}$, $k = \frac{6c\cos\varphi}{\sqrt{3}(3+\sin\varphi)}$ are for the inner circumcircle DP. The DP criterion, which shares the same parameters (c , φ) as MC, has also been widely used in the theoretical study of rock strength and suffers from the same issue of reliability as explained earlier. Moreover, as the DP criterion is a standard circle on the π plane, it leads to a large error when calculating the rock mass strength.

2.2. HB and GZZ strength criteria

2.2.1. HB strength criterion

The HB criterion was originally developed for rock masses through a large number of triaxial tests and engineering field test data, which is expressed as follows (Hoek et al., 2002):

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_b \frac{\sigma_3}{\sigma_c} + s \right)^a \quad (4)$$

where m_b , s , and a are the strength parameters of rock masses.

The HB and GZZ strength criteria used in this paper are based on Hoek's parameter system, which was established based on a large

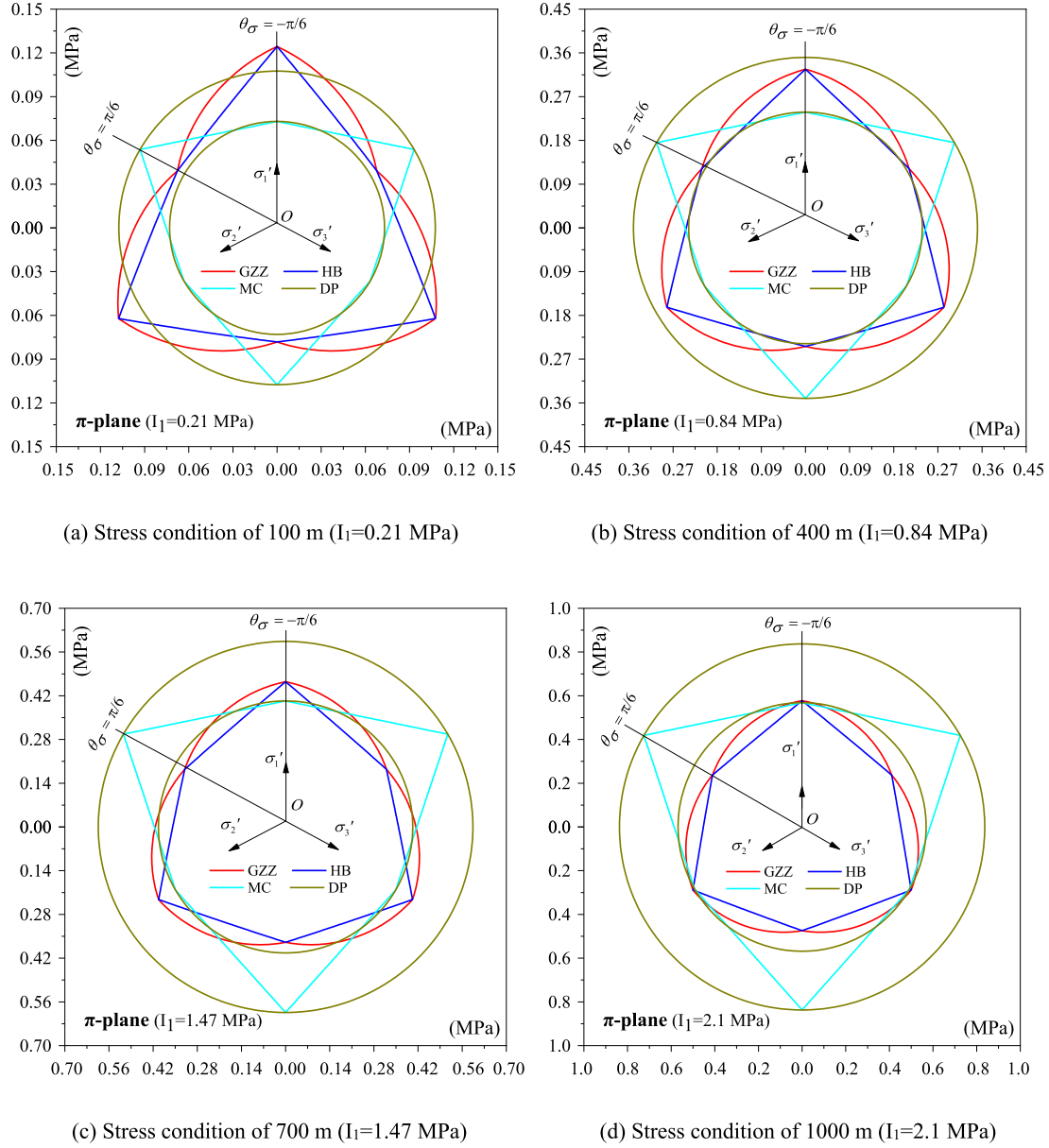


Fig. 4. Geometrical characteristics of strength criteria on the π plane under various stress levels.

number of engineering experiences and could be comprehensively obtained by performing triaxial compression tests and examining mineral compositions, discontinuity distributions, and geological features (Hoek and Brown, 2019). The geological strength indicator (GSI) is the most important parameter of HB-based strength criteria (Hoek et al., 2002), which comprehensively reflects information related to engineering hydrogeology and rock mass.

The translation relation of m_b , s , a , m_i , GSI, and D can be expressed as follows:

$$\begin{cases} m_b = m_i \exp\left(\frac{GSI - 100}{28 - 14D}\right) \\ s = \exp\left(\frac{GSI - 100}{9 - 3D}\right) \\ a = \frac{1}{2} + \frac{1}{6}[\exp(-GSI/15) - \exp(-20/3)] \end{cases} \quad (5)$$

This parameter system of HB strength criterion and its selection of value have been widely adopted and verified by many rock mass

engineers and structural designers through practical engineering applications (Hoek and Brown, 2019).

2.2.2. GZZ strength criterion

The widely used 2D HB criterion has the potential to result in inaccuracies due to lack of consideration of σ_2 . Therefore, many scholars have developed 3D HB criteria (Pan and Hudson, 1988; Priest, 2005; Zhang and Zhu, 2007; Zhang, 2008; Jiang and Xie, 2012). Based on the foundations of HB criterion and Mogi–Coulomb strength criterion (Mogi, 1967, 2006), Zhang & Zhu (Zhang and Zhu, 2007; Zhang, 2008) proposed a 3D HB criterion that combined the HB criterion with the Mogi criterion to yield the GZZ strength criterion as follows:

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right)^{1/a} + \frac{m}{2} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right) - m \frac{\sigma_1 + \sigma_3}{2} - s \sigma_c = 0 \quad (6)$$

The GZZ strength criterion can also be expressed in the form of lode angle (θ_σ) and stress invariants as:

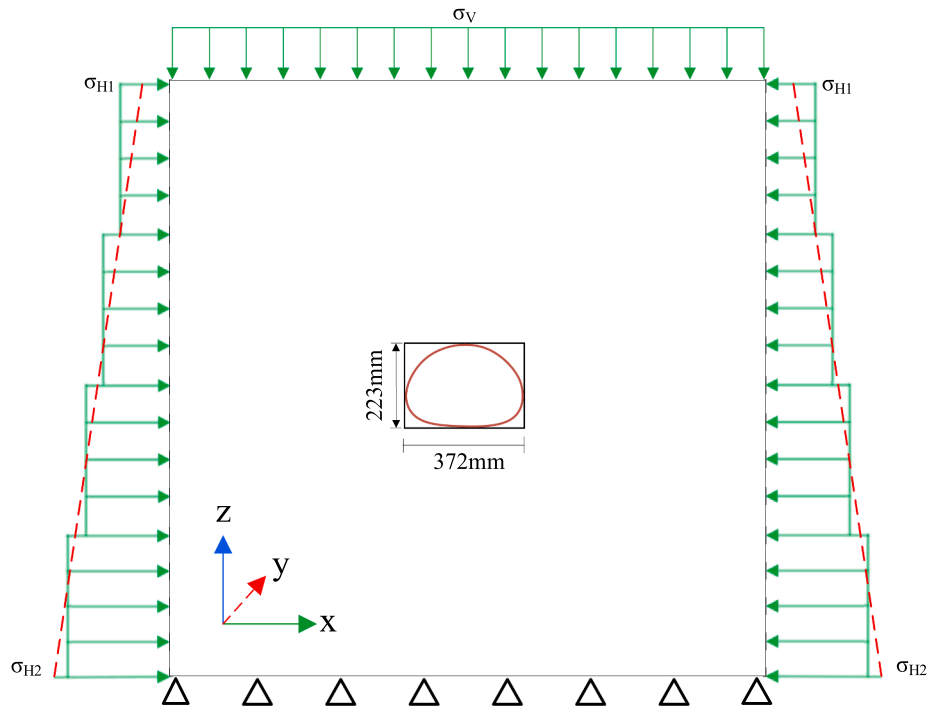


Fig. 5. Load boundary conditions of physical model test.

Table 3
Loading boundaries of physical model tests under different buried depth.

Model test process	Simulated Buried depth (m)	Applied stress/In situ stress (MPa)	σ_{H1} (MPa)	σ_{H2} (MPa)	σ_v (MPa)
1	100	0.07/2.1	0.05	0.092	0.05
2	400	0.28/8.4	0.26	0.302	0.26
3	750	0.525/15.75	0.505	0.547	0.505
4	1000	0.7/21	0.68	0.722	0.68
5	1250	0.875/26.25	0.855	0.897	0.855

Table 4
Vertical crown settlement displacements obtained from physical and numerical results (Cai et al., 2022c).

Simulated depth (m)	Model test results (mm)	Numerical results of different strength criteria			
		MC (mm)	HB (mm)	DP (mm)	GZZ (mm)
100	0.72	0.44	0.45	0.83	0.83
400	3.8	6.47	6.43	3.33	4.00
750	17.18	22.09	21.13	6.25	13.92
1,000	31.0	45.82	32.87	8.34	28.06
1,250	46.1	67.10	48.01	10.43	50.39
Relative error	none	46.4%	28.5%	48.3%	12.1%

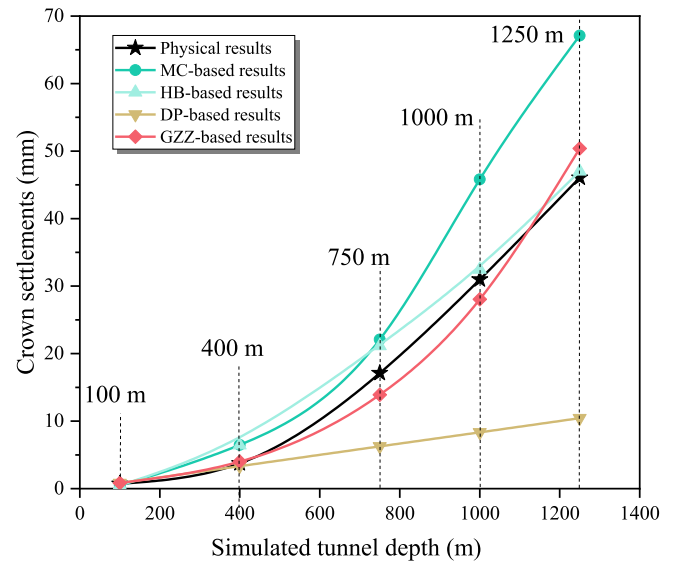


Fig. 6. Crown settlements obtained from physical model test and different strength criteria-based numerical results (Cai et al., 2022c).

concerning the non-smoothness and non-convexity of the original one (Eq. (7)). Its mathematical expression is:

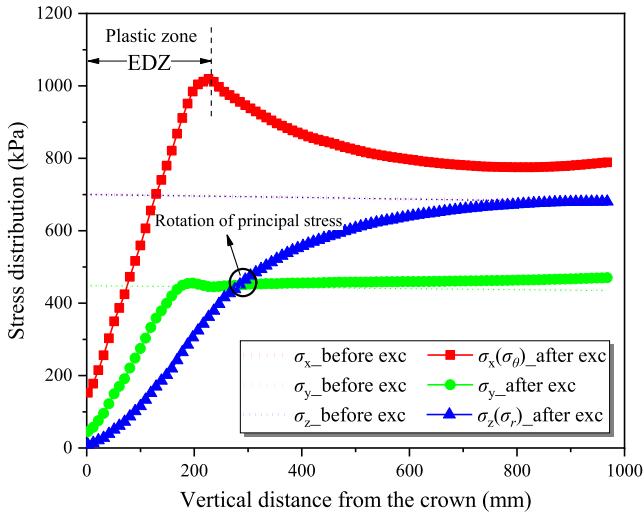
$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3J_2} \right)^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - \sigma_c = 0 \quad (8)$$

GZZ (smooth GZZ) is originally developed from HB criterion and Mogi criterion with a nonlinear envelope for rock masses, and it also can be used for deep rock mass that experience high and complicated geostresses in the 3D and nonlinear system (Zhu et al., 2023).

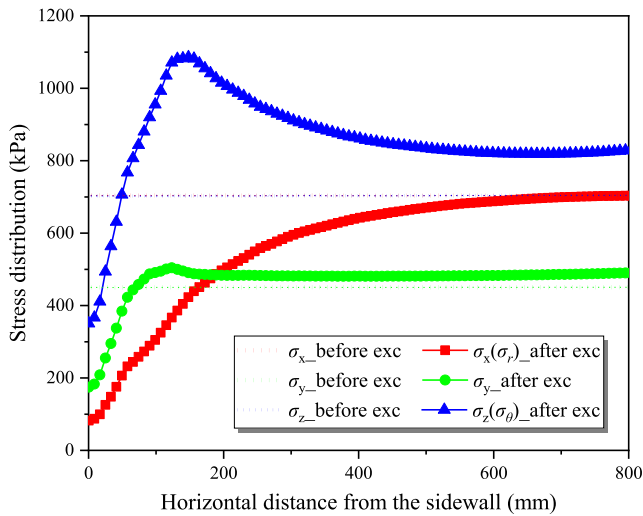
$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3J_2} \right)^{1/a} + \left(\frac{3 + 2\sin\theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - \sigma_c = 0 \quad (7)$$

Fig. 1 shows the GZZ and HB strength criteria on the π plane. The intersection points of the two criteria is $\theta_\sigma = \pm \pi/6$ (i.e. triaxial compression stress state and triaxial tensile stress state), which ensures that the parameters from Hoek's recommendation (i.e. Eq. (5)) can be used for GZZ when calculating the strength stability of rocks and rock masses (Zhang and Zhu, 2007; Zhang, 2008).

Cai et al. (2021) developed a smooth GZZ strength criterion in terms of stress invariants with smooth and derivable geometry on the π plane



(a) Stress distribution of crown before and after excavation



(b) Stress distribution of sidewalls before and after excavation

Fig. 7. Stress distribution of surrounding rock before and after excavation.

3. Excavation unloading of physical model tunnels under various buried depths

3.1. Physical model tunnels and calculation of strength parameters

Several physical model tests were conducted to simulate and investigate the progressive characteristics of deformation and failure of surrounding rock. The physical test settings and crown-displacement monitoring location are shown in Fig. 2 (Vu, 2014; Vu et al., 2016). In order to simulate a practical tunnel with a size of 11.2×6.7 m, the size of the model test equipment of $2 \times 2 \times 0.4$ m with a similarity ratio of 30 was selected, and the profile of model tunnel unloading was 372×223 mm.

The uniaxial compressive strength of the model rock (σ_{ci}) is 51 kPa (0.051 MPa). It is difficult to calculate GSI by using the field acquisition method because of the homogeneity of the material. Therefore, the basic physical and mechanical parameters can only be obtained from the conventional triaxial test data of model rocks; the fitting results are shown in Fig. 3.

The strength parameters (i.e., cohesion c and internal friction angle φ) of MC can be obtained by linear fitting of the strength data. However, there are two main methods for calculating the parameters of the HB strength criterion (m, s, a), i.e., full-fitting (Approach #1) and correlate-fitting (Approach #2). Full-fitting considers parameters m, s , and a as HB-criterion parameters to fully fit the rock-strength data using the optimization method. Correlate-fitting considers m, s , and a as functions of GSI and D ($D = 0$ for indoor test) to fit the data by using Eq. (5) and then subsequently obtaining parameters m, s , and a by reusing the same equation. The results based on full-fitting and correlate-fitting are summarized in Table 1.

It can be concluded from Table 1 that the value of m obtained by full-fitting is significantly lower than that of the recommended value of sandstone (Hoek and Brown, 1997). Although the accuracy of full-fitting is higher than that of correlate-fitting, the physical meaning and range of parameters cannot be considered. The GSI obtained through correlate-fitting is 100.46, which is closer to the GSI of 100 recommended by Hoek and Brown (Hoek and Brown, 1997) for intact rocks with no evident structural plane. The value of GSI obtained using correlate-fitting reflects the parameters of intact rock, which is in accordance with the nature of the produced model rock, whose homogeneity and integrity are much better than those of the practical rock. Thus, $GSI = 100$ was believed to be reliable for the calculation, and correlate-fitting was treated as more reliable compared with full-fitting. Notably, the structure of engineering rock mass is complex, and full-fitting cannot consider the physical and mechanical meanings of rock parameters and the internal relationships of parameters because of the pure mathematical fitting. Thus, the rock parameters obtained through back analysis or fitting using full-fitting are bound to be considerably different from the actual strength parameters. Conversely, correlate-fitting can consider the physical meaning of rock parameters and the internal relationship between parameters by treating strength parameters as functions of GSI, which can then be used for positive analysis of geotechnical parameters. The analysis results of the two approaches are consistent with the research conclusions in the literature (Hoek et al., 2002; Alejano et al., 2017). The corresponding calculation parameters are shown in Table 2.

3.2. Geometrical characteristics of strength criteria under various stress levels

In the 2D stress space (σ_1 - σ_3), the plot of the rock strength (σ_1) with different confining pressures (σ_3) forms the strength envelope, similar to the MC and HB strength criteria shown in Fig. 3. For the 3D strength criterion, the curve of the strength envelope can be illustrated on the π plane, and the envelope varies with stress levels (I_1). Fig. 4(a), (b), (c), and (d) are the strength envelopes of different strength criteria on the π plane when values of I_1 are 0.21, 0.84, 1.47, and 2.1 MPa, corresponding to the buried depths of 100, 400, 700, and 1000 m, respectively, in practical in situ stress levels.

It can be concluded from Fig. 4 that the yield surface gradually extends outward with increase in I_1 and buried depth on the π plane. Correspondingly, the rock strength, σ_1 , gradually increases with increase in σ_3 in the 2D stress space (σ_1 - σ_3). The DP criterion is a circle (circumscribed and inscribed circle of MC) on the π plane, and the stress radius of MC in its triaxial compression stress state ($\theta_\sigma = -\pi/6$) is clearly shorter than that of the triaxial tension stress state ($\theta_\sigma = \pi/6$), which is inconsistent with the true triaxial strength characteristics of most rocks (Mogi, 1967, 1971; Al-Ajmi and Zimmerman, 2005). The strength envelopes of HB and GZZ are outside of DP and MC at $\theta_\sigma = -\pi/6$ when I_1 is small, whereas they gradually transfer to the inside of DP and MC with an increase in I_1 . The results show that the yield stresses of HB and GZZ are higher than those of MC and DP under shallow depths with low I_1 , but they are lower than those of MC and DP under greater depths with high I_1 . It can be inferred that HB and GZZ will be located completely inside the MC and DP envelopes when the buried

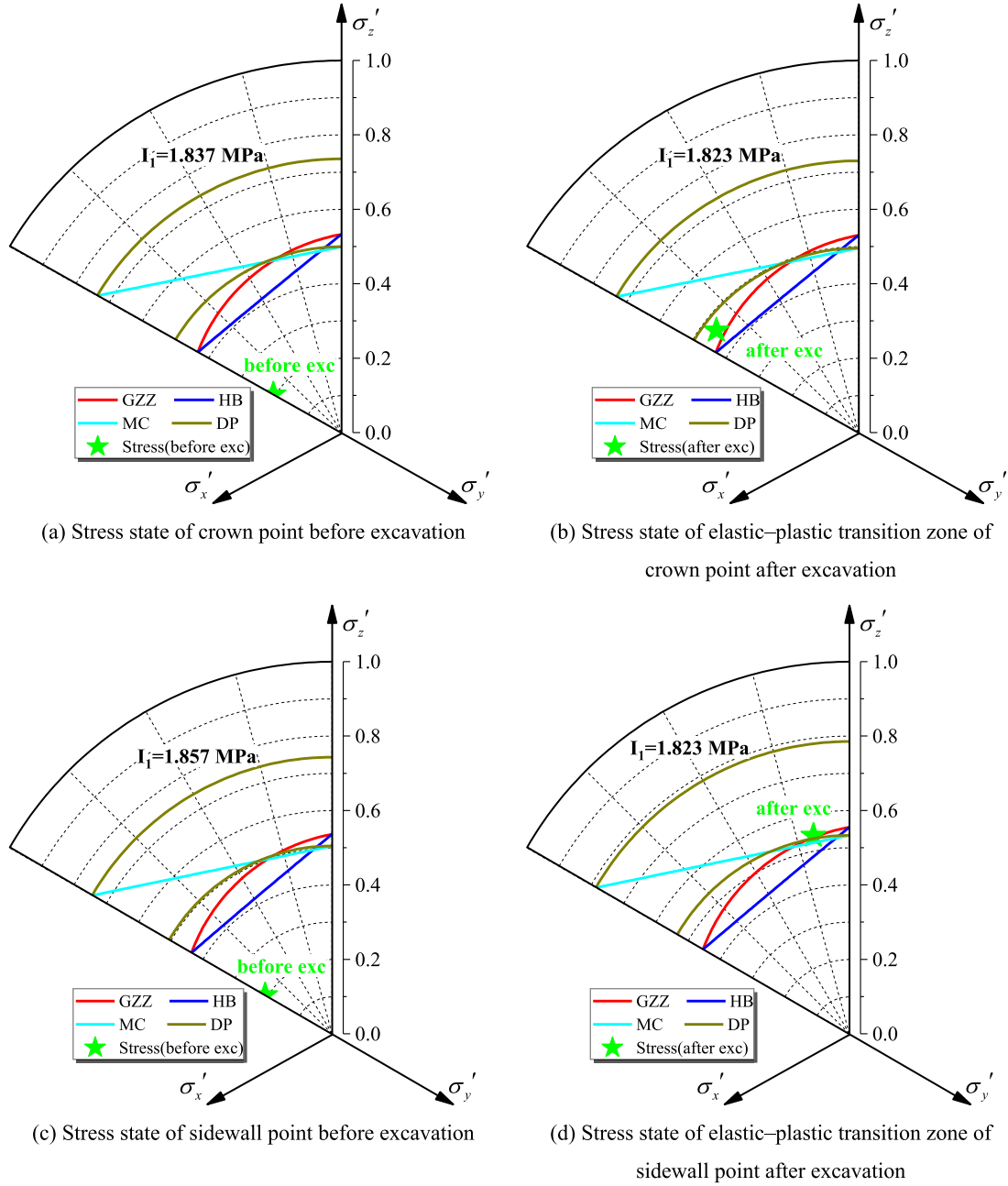


Fig. 8. Stress state of surrounding rock before and after excavation.

depth exceeds 1000 m, which indicates that the influence of confining pressure on rock strength is related to the stress level. This characteristic on the π plane is in accordance with the 2D criterion in the space of σ_1 - σ_3 , similar to MC and HB as shown in Fig. 3. The HB envelope is located above MC at low σ_3 but gradually transfers to the location below MC with an increase in σ_3 . This nonlinearity of 2D HB better reflects the nonlinear variation of rock strength with confining pressure, compared to linear MC. Based on the above discussion, it can be concluded that the envelope of GZZ on the π plane can reflect the nonlinear relationship between rock (rock mass) strength and stress level (I_1).

3.3. Comparison of monitor data with numerical results of crown settlement

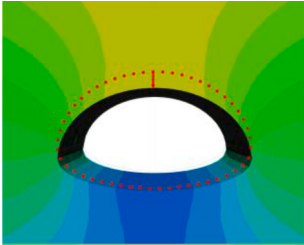
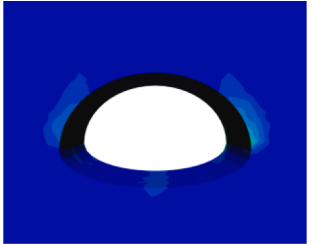
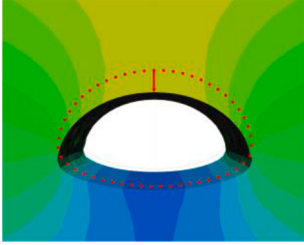
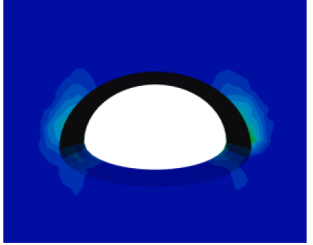
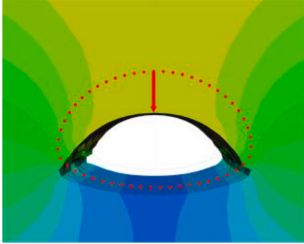
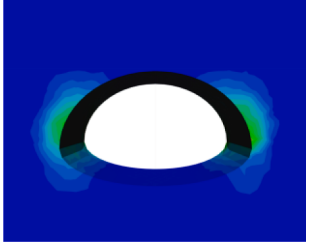
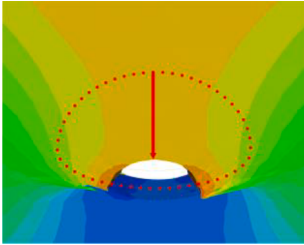
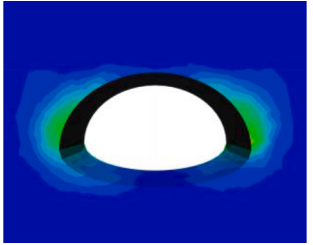
A 3D finite element model was established according to the settings of the physical model. The two sides and top boundaries and were

maintained with synchronous and equal pressure to simulate a certain lateral pressure coefficient ($k = \sigma_x/\sigma_z$), and other boundaries (i.e., bottom, back and front boundaries) were fixed in the normal directions. In the calculations of the model tunnels unloading under various buried depths, MC, DP, HB, and GZZ strength criteria were used for calculation and comparison.

The boundary condition loading of the model test is shown in Fig. 5. The depths of 100 m to 1250 m of the model tunnel were selected, as shown in Table 3. The σ_v (vertical loading) simulated the different buried depths of tunnels, and a trapezoidal load was symmetrically applied to simulate a certain k (lateral coefficient). The model test process was divided into five steps (Table 3).

Table 4 and Fig. 6 illustrate the calculated results of the crown settlement displacements for $k = 1.0$ under various depths and different strength criteria. Evidently, the MC-based numerical results are larger than the monitor values, which indicates the prediction of the 2D MC

Table 5Physical and numerical results (GZZ) for $k = 0.25$.

Depth(m)	Results of model test	Total displacement	Maximum plastic shear strain
150			
200			
400			
700			

criterion is too conservative because of the lack of consideration of σ_2 . The 3D DP-based results of the crown settlement present a linear relation to buried depth, which is significantly lower than the monitor values. Although the σ_2 on rock strength is considered in DP, it also exists considerable difference from the practical strength of rock mass because of the geometric form of a circle on the π plane, which results in optimistic results of tunnel deformation. The results of GZZ, which directly use the HB's parameters, are in good agreement with the monitoring deformation data, and its mean relative error is 12%, which obtain the highest overall accuracy among these four strength criteria. The results of HB under 1000 and 1250 m buried depths are close to GZZ-based results as well as the monitoring values.

The comparison results of different strength criteria indicate that a reliable 3D strength criterion, such as the GZZ criterion, should be adopted for deep rock tunnels under complicated stress conditions. The above discussion also shows that the direct use of the same parameters for GZZ as HB is reliable and accurate. More detailed discussions were provided by Cai et al. (2022c).

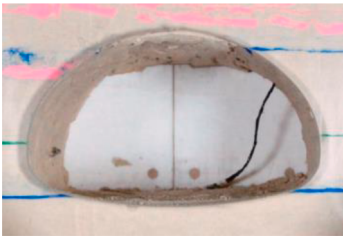
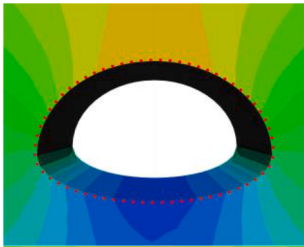
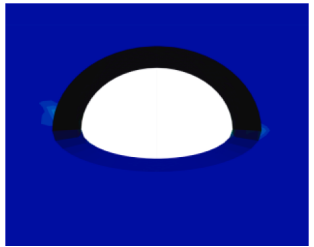
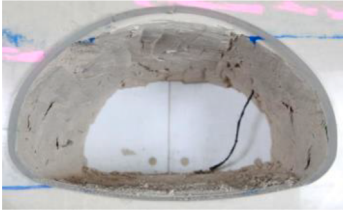
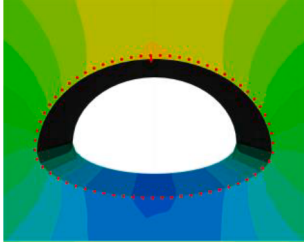
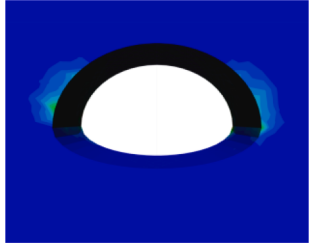
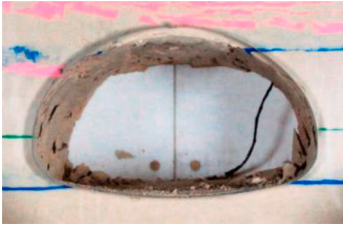
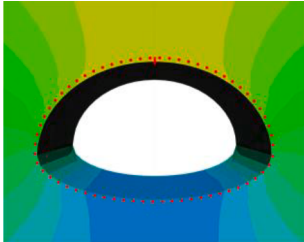
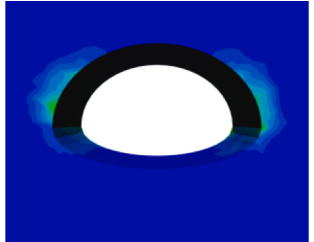
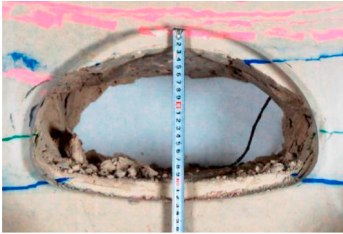
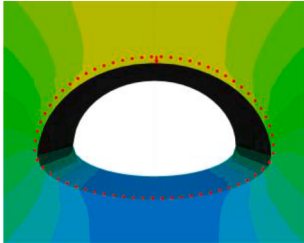
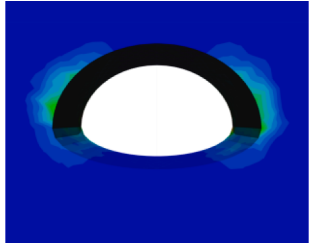
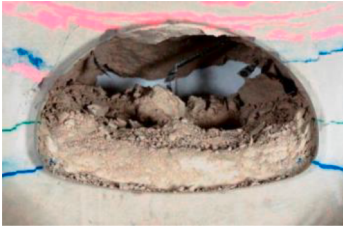
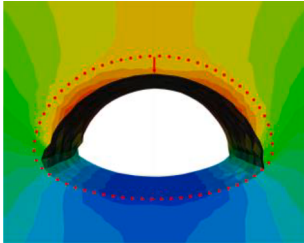
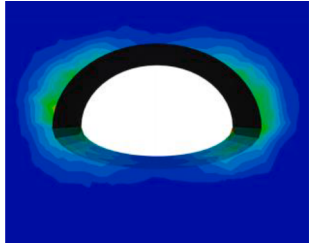
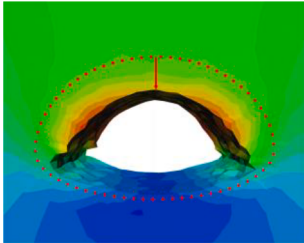
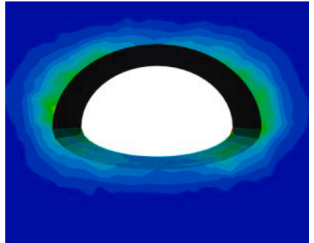
3.4. Stress distribution of surrounding rock before and after excavation for 1000-m-buried tunnel with $k = 1.0$

Fig. 7(a) and (b) show the stress distributions of crown and sidewall calculated using GZZ, respectively. The original stress state of rock is $k = 1.0$ —the stress in the x direction equals that in the z direction, and the

stress in the y direction is the automatic equilibrium geostress. Excavation unloading causes a redistribution of surrounding rock stresses. The location of tangential stress is 220-mm vertically from the crown and 170-mm horizontally from the sidewall, respectively, within which the surrounding rock is in the plastic zone with poor stability. The principal stress axis is transferred in both time and space distributions: (1) The horizontal tangential stress $\sigma_x(\sigma_\theta)$ of the crown before and after excavation is σ_1 , and σ_y is transferred from σ_3 before excavation to σ_2 in the plastic zone after excavation. The stress in the elastic zone remains unchanged. (2) σ_1 , σ_2 , and σ_3 in the plastic zone of the crown are σ_x , σ_y , and σ_z , respectively, whereas σ_1 , σ_2 , and σ_3 in the elastic zone (250-m vertically away from the crown) are σ_x , σ_z , and σ_y , respectively. Similarly, the principal stress axis near the sidewall also has an evident transfer in time and space, and the transfer location is extremely close or slightly after the corresponding position at the peak tangential stress ($\sigma_{\theta, \max}$).

The critical range of the excavation damage zone corresponds to the position of the maximum tangential stress $\sigma_{\theta, \max}$ around the tunnel, and the stress conditions at this point before and after excavation are shown in Fig. 8(a)–(c). The original stresses of the rock at the crown and sidewall are both located within the strength envelope on the π plane, which are also in the elastic stress states. The stress condition of the excavation damage zone (EDZ) position is located at $\theta_\sigma = -\pi/6$ for the crown and $\theta_\sigma = \pi/6$ for the sidewall on the π plane. Additionally, these two points are located on the GZZ strength envelope, indicating that σ_θ ,

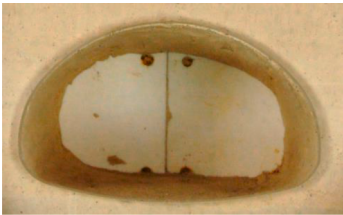
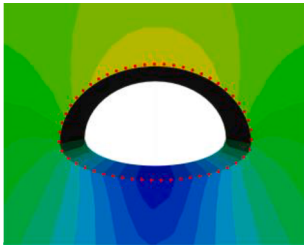
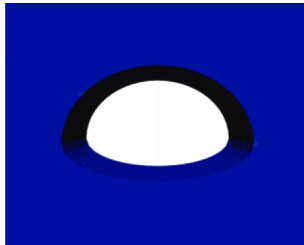
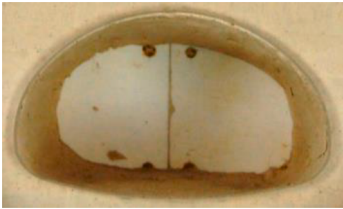
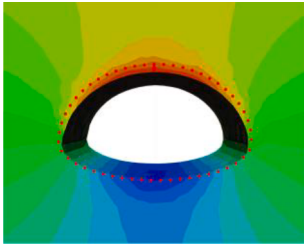
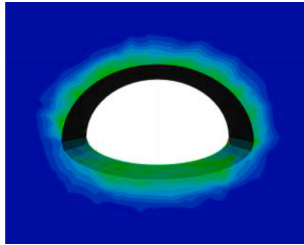
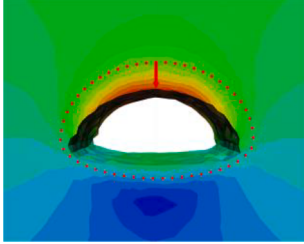
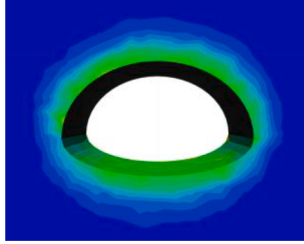
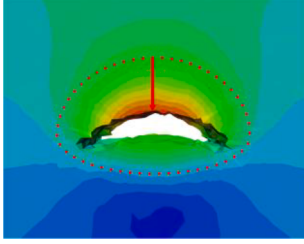
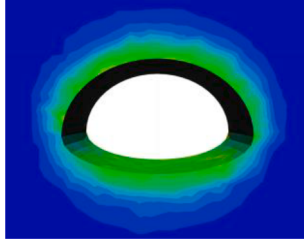
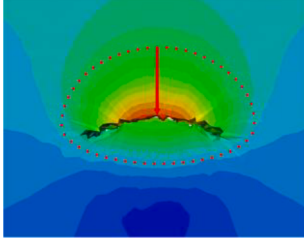
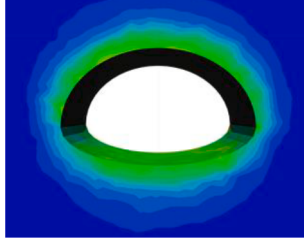
Table 6
Physical and numerical results (GZZ) for $k = 0.5$.

Depth(m)	Results of model test	Total displacement	Maximum plastic shear strain
100			
200			
250			
300			
700			
1000			

$\sigma_{\max}$ is in the elastic–plastic transition zone. For both the crown and sidewall, the stress conditions before and after excavation are in the state of $\theta_\sigma = -\pi/6$ or $\theta_\sigma = \pi/6$, where GZZ intersects HB on the π plane. This may qualitatively explain why the calculation results of GZZ are similar to HB under 1000- and 1250-m-buried rock tunnels when $k = 1.0$ (Fig. 6). Thus, the results obtained using the 3D strength criterion

are similar to those of the 2D strength criterion when the rocks are in pseudo triaxial stress conditions. For deep tunnels under true triaxial stress conditions (i.e., $\sigma_1 > \sigma_2 > \sigma_3$), reliable results rely on the utilization of the 3D strength criterion.

Table 7Physical and numerical results (GZZ) for $k = 1.0$.

Depth(m)	Results of model test	Total displacement	Maximum plastic shear strain
100			
400			
750			
1000			
1250			

4. Deformation and failure of model tunnels under complicated stresses

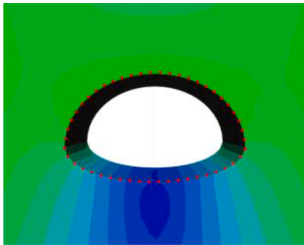
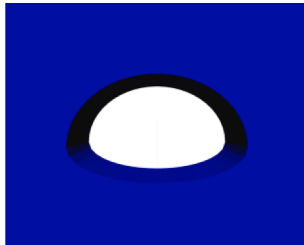
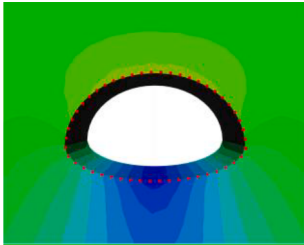
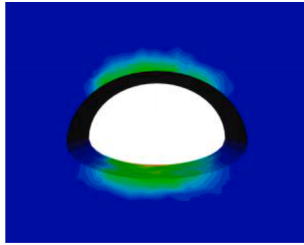
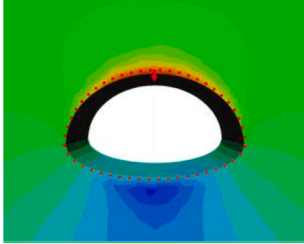
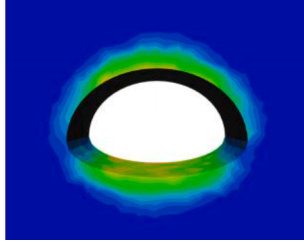
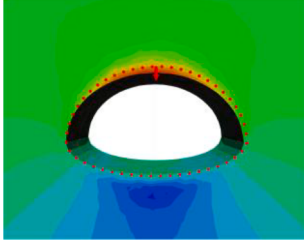
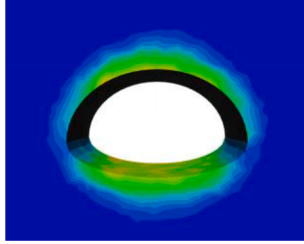
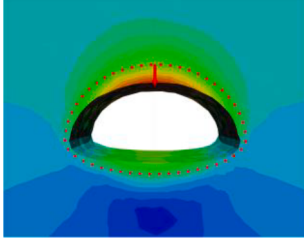
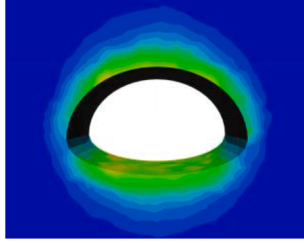
Various coefficients of horizontal pressure k can be used to simulate the complicated stress environment of the in situ geological state. Here, k equals σ_x/σ_z , and σ_y is obtained by the automatic balancing of elastic-plastic stresses according to the numerical results of GZZ. Thus, the rock of the physical model test is indeed under true triaxial stress conditions (i.e., $\sigma_1 > \sigma_2 > \sigma_3$).

4.1. Deformation and failure laws of model tunnels excavation for $k = 0.25$

Table 5 summarizes the results of physical model tests and the GZZ-based numerical calculations at different buried depths for $k = 0.25$. The rock support against the pressure on the sidewall is weak because of the

small value of k . An evident block-drop failure phenomenon appears at the sidewalls when the tunnel depth is 200 m. The crown settlement is considerably large because of the vertical principal stress, $\sigma_2(\sigma_1)$, and the tunnel is nearly closed at 700 m because of the large vertical deformation. The GZZ-based calculation also shows that the concentrated plastic zone first appears on the sidewalls with an increase in buried depth, and the crown area is still in the elastic state even when the tunnel is closed and totally damaged. Thus, the failure of the sidewalls is the main reason for the tunnel close and crash for $k = 0.25$. The deformation and failure process can be summarized as follows: convergent deformation; micro-cracks appearing at sidewalls; spalling at sidewalls; sidewalls shear fractures; failure at crown; and, finally, collapse.

Table 8
Physical and numerical results (GZZ) for $k = 1.25$.

Depth(m)	Results of model test	Total displacement	Maximum plastic shear strain
100			
150			
350			
400			
700			

4.2. Deformation and failure laws of model tunnels excavation for $k = 0.5$

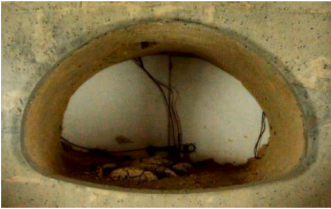
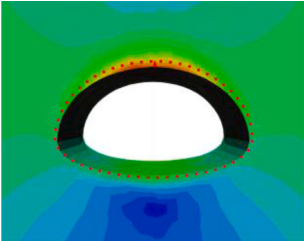
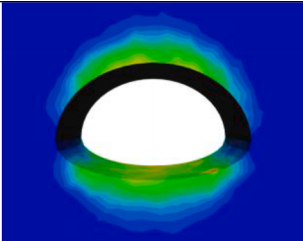
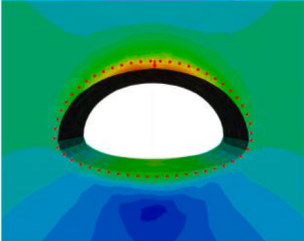
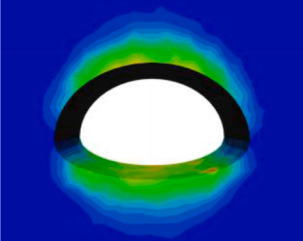
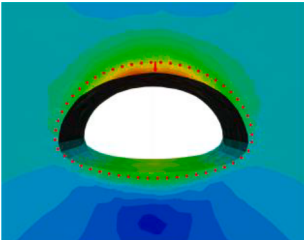
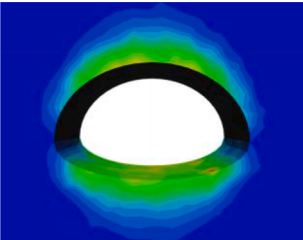
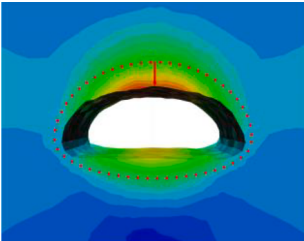
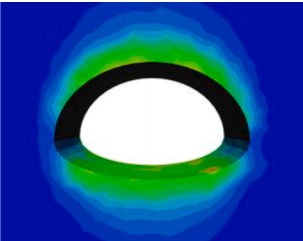
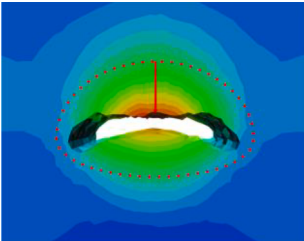
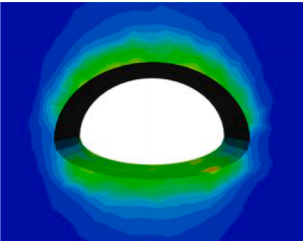
Table 6 lists the results of deformation and failure of the model test and numerical results using GZZ at different buried depths for $k = 0.5$. Similar to the case of $k = 0.25$, the horizontal lateral stress for $k = 0.5$ is still small, and there is an evident drop at the sidewalls when tunnel depth is 300 m. The drop at sidewalls of 300–700 m induces an evident settlement deformation of the crown. The numerical analysis results show that the concentrated plastic zone first appears at the sidewalls with an increase in buried depth, and the local plastic zone also appears at the crown in its deep formation. The stability of the tunnel for $k = 0.5$ is still dominated by the sidewalls' failure, and the deformation and failure process can be summarized as follows: convergent deformation; microcracks at sidewalls; spalling at sidewalls; sidewalls shear fractures with some block drops; crown failure and remote crack formation; and,

finally, collapse.

4.3. Deformation and failure laws of model tunnels excavation for $k = 1.0$

Progressive deformation and failure laws of the physical model test results with GZZ-based numerical calculations for $k = 1.0$ at different tunnel depths are compared and listed in Table 7. The model tunnel collapses in a large area at a depth of 1250 m, and the tunnel is nearly closed at 1250 m according to the numerical calculation. The plastic zone is evenly distributed around the tunnel with an increase in buried depth, and the overall shear failure of the crown and sidewalls constitute the failure mechanism of the tunnel for $k = 1.0$. The unsupported tunnel failure for $k = 1.0$ shows a progressive process which could be summarized as: crown settlement and horizontal convergence; cracks and block drops at the sidewalls; sidewalls shear fracture; crown failure and

Table 9
Physical and numerical results (GZZ) for $k = 1.5$.

Depth(m)	Results of model test	Total displacement	Maximum plastic shear strain
350			
400			
450			
700			
1000			

remote crack formation; and, finally, collapse.

4.4. Deformation and failure laws of model tunnels excavation for $k = 1.25$

Table 8 shows the results of the physical model test and the GZZ-based numerical results under different buried depths for $k = 1.25$ with a large horizontal lateral stress. The crown and sidewalls show an evident block drop at 350 m, and the crown drop is more serious than that of the sidewalls when the depth of tunnel is >400 m. The crown shows a certain degree of bottom uplift deformation at 700 m. The numerical analysis results show that the concentrated plastic zone first appears at the crown with an increase in buried depth, and the local plastic zone also appears at the sidewalls in the deep tunnels. The tunnel is dominated by the failure of the crown, but the overall deformation is not large. The failure process can be summarized as follows: stable

tunnel after excavation; microcracks at the sidewalls and crown; spalling at the sidewalls and crown; shear fractures at the sidewalls, shoulders, and crown; failure at the crown, and finally collapse.

4.5. Deformation and failure laws of model tunnels excavation for $k = 1.5$

Table 9 presents the results of deformation and failure of the model test and the GZZ-based numerical calculation under varying buried depths for $k = 1.5$ with a large horizontal lateral stress. The tunnel crown shows a V-shaped drop at 400 m, and it falls considerably at a depth of 700 m. Simultaneously, the local sidewalls fall until overall collapse. The numerical analysis results show that there is a concentrated plastic zone in the crown with an increase in buried depth, and the sidewalls are in the local plastic state in a limited area even when the tunnel collapses. Thus, the V-shaped collapse of the crown is considered to be the main

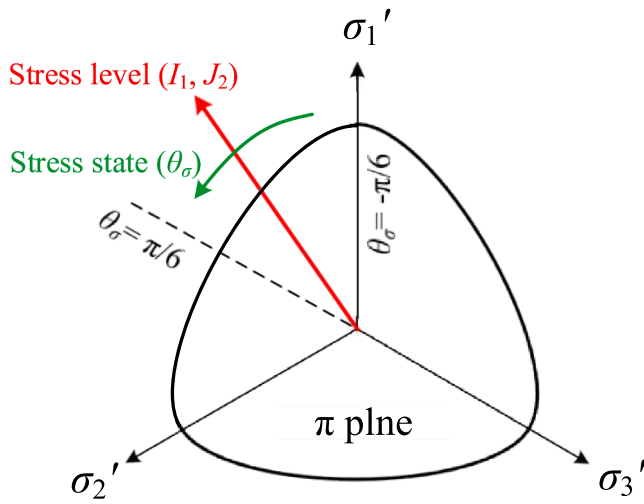


Fig. 9. Diagram of complicated stress conditions on the π plane.

failure pattern for the rock tunnel for $k = 1.5$. The deformation and failure process can be summarized as follows: stable tunnel after excavation; V-shaped deformation at the crown; deformation enlarging at the crown and microcracks at the sidewalls; and tunnel collapse.

4.6. Discussion on progressive behaviors of surrounding rocks under complicated stress conditions

It can be drawn from the above analysis that the GZZ-based numerical results are in good agreement with the model test results from the aspects of progressive deformations and failure characteristics and failure modes of surrounding rock. The physical model tests and the numerical results show that the progressive deformation and failure laws of surrounding rock are closely related to the stress condition (k) and stress level (buried depth). As shown in Fig. 9, the stress condition and stress level respectively correspond to the circumferential coordinate (θ_σ) and radial coordinate ($r = \sqrt{2J_2}$) on the π plane. The failure mechanism and failure location of surrounding rock are related to the stress condition. Overall, the deformation and failure of surrounding rock mainly occur in the direction of σ_3 around the tunnel (i.e., sidewalls for $k < 1.0$ and crown for $k > 1.0$), and the stress level determines the severity of the failure. Excavation unloading of the tunnel reduces the support of stress in the direction of the original σ_3 , where the plastic zone appears and local tensile cracking is produced at a shallow buried depth. Then, the degree of shear failure gradually increases with an increase in buried depth, which may induce the local failure of surrounding rock in the direction of σ_1 . This eventually leads to the overall shear instability of the model tunnel. For actual tunnel engineering, the deformation in the direction of the original σ_3 should be reinforced with priority to control the overall stability of a tunnel.

5. Nonlinear stress characteristics of model tunnels under various buried depths

The stresses are the internal cause of deformation and failure of the rock mass. As a case study, Fig. 10(a) and (b) reveal horizontal stress σ_x and vertical stress σ_z of tunnels with different buried depths for $k = 1.0$, where σ_x is the tangential stress of the crown $\sigma_\theta(\sigma_1)$ and σ_z is the tangential stress of the sidewalls $\sigma_\theta(\sigma_1)$.

The high ground stress or stress concentration may not cause rock yield, but the strong disturbance nature of deep rock tunnels may cause unexpected unloading, which in turn can lead to serious engineering problems. Therefore, the high ground stress area and stress concentration are potential unstable areas which should be given adequate attention. It can be observed from Fig. 10(a) and (b) that the maximum

principal stresses (σ_θ) for the crown and sidewalls of the shallow-buried tunnel are located on the free face of the tunnel (the blue region in Fig. 10(a) and (b)), such as the crown at 100 m in Fig. 10(a) and the sidewalls at 100 and 300 m in Fig. 10(b). The stress concentration area (i.e., the EDZ) gradually transfers far away from the tunnel free face along the radial direction with an increase in buried depth. The stress concentration area is also the area where tunnel disasters frequently occur for deep rock masses (e.g., rock burst). The stress analysis results indicate that the rock burst in a shallow tunnel mainly occurs near the tunnel, but the damage and rock burst originate far away from the surrounding rock at a certain distance with an increase in buried depth. Rock burst disasters become somewhat hidden in deep rock tunnels. The analysis results from the GZZ-based calculation coincide with the conclusions found in the literature (Ma et al., 2015; Feng et al., 2019; Gong et al., 2019).

The previous analysis showed that $\sigma_{\theta, \max}$ can be considered as the EDZ at the elastic–plastic transition point, where the principal stress axis is significantly transferred in time and space. Fig. 11 and Fig. 12 show the θ_σ of the crown and sidewalls under buried depths for different values of k . The red curves in the figure represent $\sigma_{\theta, \max}$ (i.e., the EDZ or elastic–plastic transition zone). For the crown and sidewalls, the distance of the position of $\sigma_{\theta, \max}$ to the free face of the tunnel increases approximately linearly with an increase in buried depth, and it can be inferred that the area of the plastic zone is proportional to the square of the buried depth. Thus, there is a significant difference in the distribution of plastic zone and the transfer phenomenon of the principal stress axis between deep- and shallow-buried tunnels before and after excavation. The difference law and degree of difference are approximately linearly correlated with the buried depth. It can be considered that the characteristics of stress distribution and stress level are the core factors responsible for deformation and failure of deep tunnels.

6. Nonlinear deformation characteristics of model tunnels under various buried depths

Fig. 13(a) and (b) present the displacement curves of the crown settlement and the horizontal deformation of the right sidewall of modeled tunnels under various buried depths at different values of k . The results show that the crown settlement and the horizontal deformation around the tunnel present nonlinear increasing trends with an increase in buried depths, and the deformation of shallow tunnels can be described by a linear relationship. However, the deep tunnels reflect significant nonlinear characteristics. The nonlinear characteristic is closely related to the stress condition (k); the larger the horizontal stress (the larger k), the more evident is the nonlinear characteristic.

Linear function ($y = ax + b$), parabolic function ($y = ax^2 + b$), exponential function ($y = ae^{bx}$), and hyperbolic catenary function ($y = a \cosh(bx)$) are used to fit the nonlinear curves of Fig. 13(a) and (b), and the results are shown in Table 10. The fitting accuracy of linear function is the lowest; the fitting accuracy of exponential function is the highest (average $R^2 = 0.994$), which is slightly higher than that of hyperbolic catenary function (average $R^2 = 0.993$), among which the hyperbolic catenary function has been widely used in engineering practice.

For a rock tunnel without support, the deformation of surrounding rock has significant nonlinear characteristics with an increase in buried depth, which can be described by the divergent exponential or hyperbolic catenary function. The nonlinearity is directly related to the stress conditions; the larger the horizontal tectonic stress, the more evident are the nonlinear characteristics. The deformation characteristics of a shallow tunnel can be described with an approximate linear law, which is not applicable for deep-buried rock tunnels, which show nonlinear divergence characteristics. Thus, the design theories and practical experiences collected from shallow tunnel engineering, which is dominated by self-weight stress, are not applicable for deep tunnels.

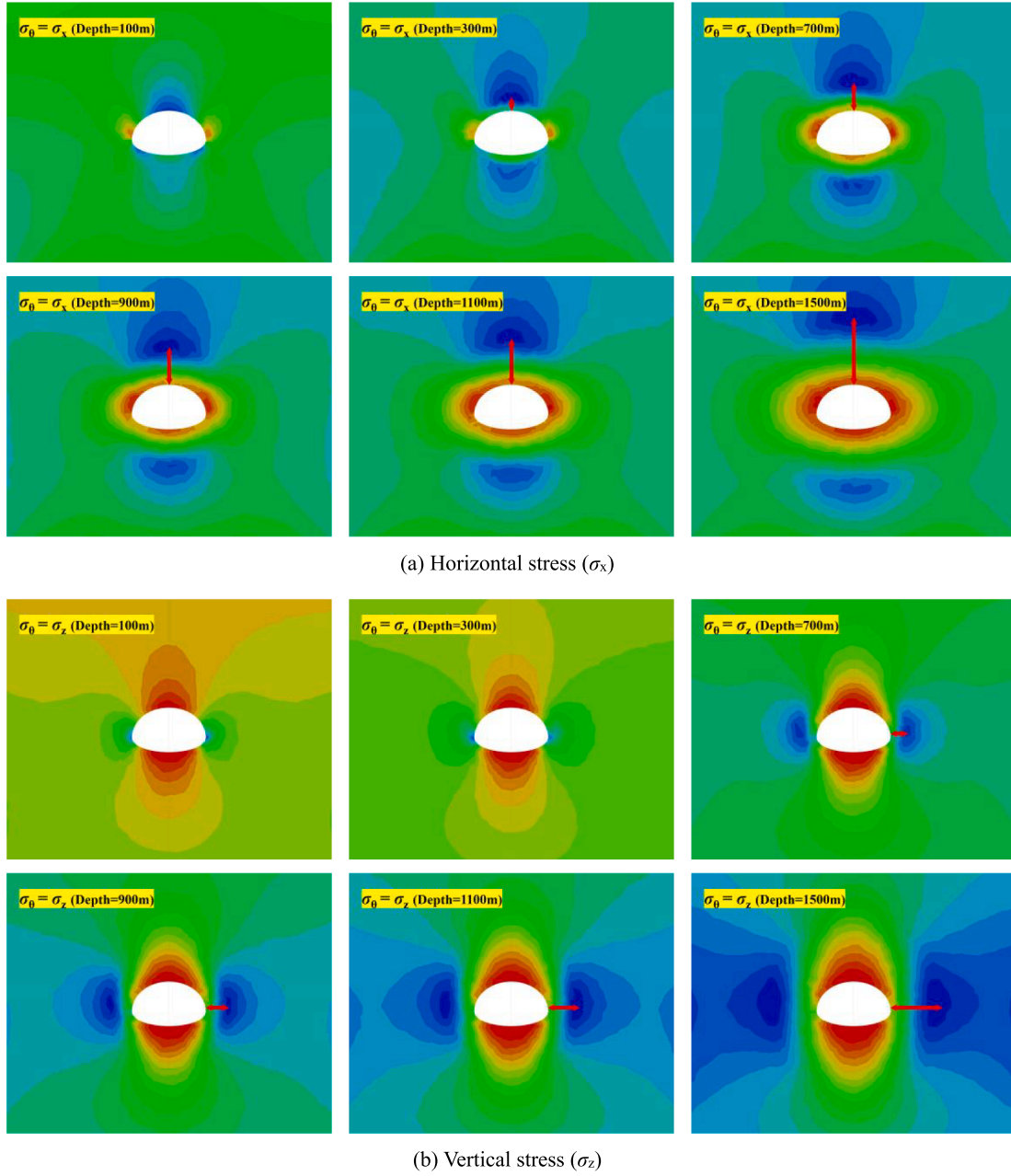


Fig. 10. Stress distribution of tunnels under different buried depths for $k = 1.0$.

7. Conclusions

In this study, physical model tests and GZZ-based numerical calculation of model tunnels under complicated stress conditions and various buried depths were carried out to examine the progressive deformation and failure laws of unsupported tunnels, and the following conclusions were drawn:

- (1) Considering the strength parameters as functions of GSI (Eq.) to obtain the GSI value by fitting the rock-strength data and for calculating the corresponding values of m , s , and a is reliable and meaningful. The obtained strength parameters of HB can then be directly used as the strength parameters for GZZ. Comparisons between the monitoring results of physical model tests with the calculation results using different criteria show that the GZZ-based calculation results showed the best agreement with the monitoring data, and GZZ strength criterion should be adopted

for deep rock tunnels under complicated stress conditions ($\sigma_1 > \sigma_2 > \sigma_3$).

- (2) The physical model tests and the GZZ-based numerical results show that the progressive deformation and failure laws of surrounding rock were closely related to the stress condition (k) and stress level (buried depth). It was primarily the sidewalls that failed under the gravity stress-dominated field ($k < 1.0$), and the crown failed under the horizontal tectonic stress-dominated field ($k > 1.0$). The failure mode changed from tensile failure in shallow-buried tunnels to shear failure in deep-buried tunnels.
- (3) The distance of the position of $\sigma_{\theta, \max}$ to the free face of the tunnel increased approximately linearly with an increase in buried depth. The principal stress axis transfers in time (before and after excavation) and space (plastic zone and elastic zone) with an increase in buried depth, and a larger depth causes a wider transfer range. It can be considered that the divergences of distribution of the elastic-plastic zone and the transfer of the

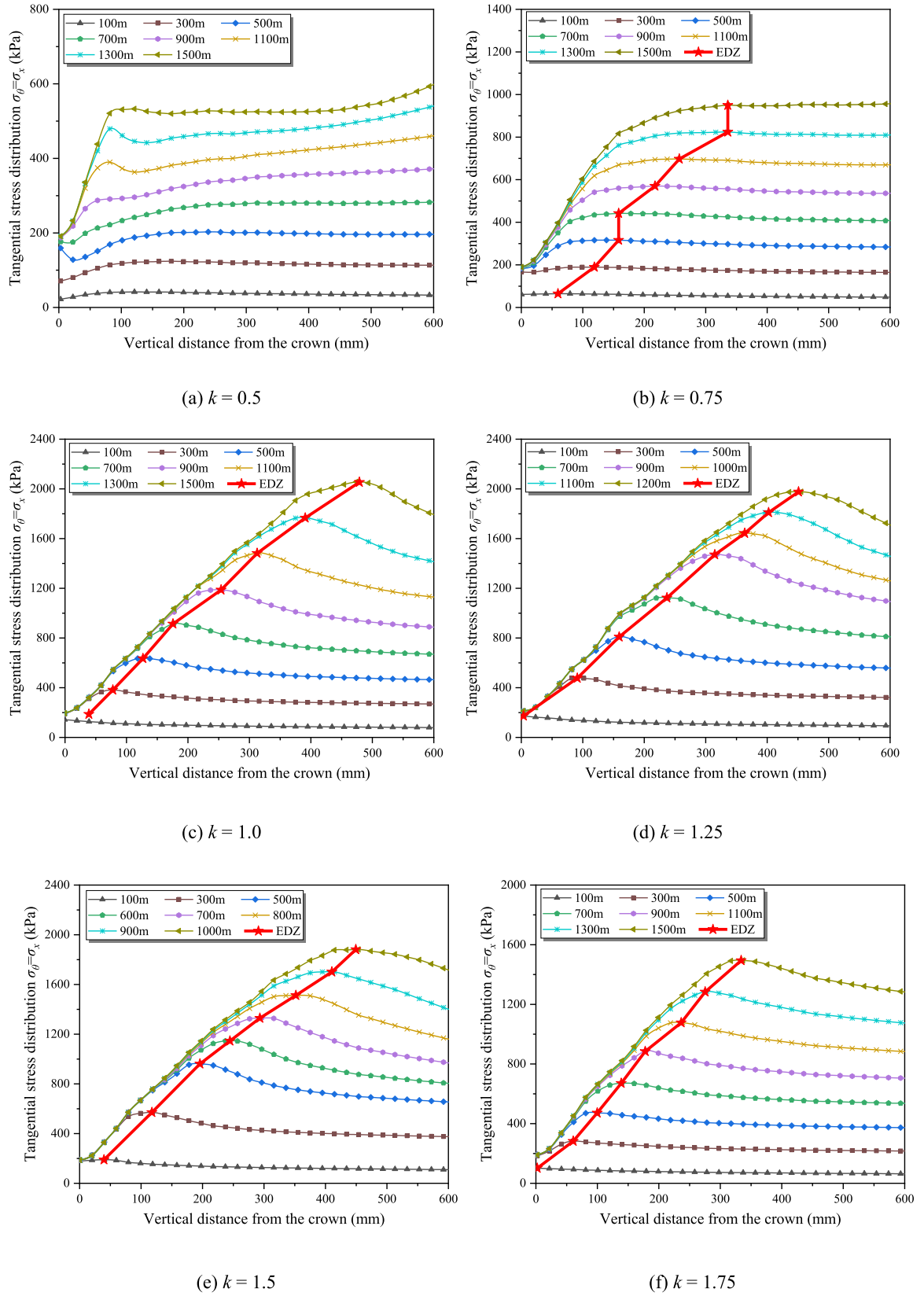


Fig. 11. Distribution of tangential stress vertically from crown for various values of k .

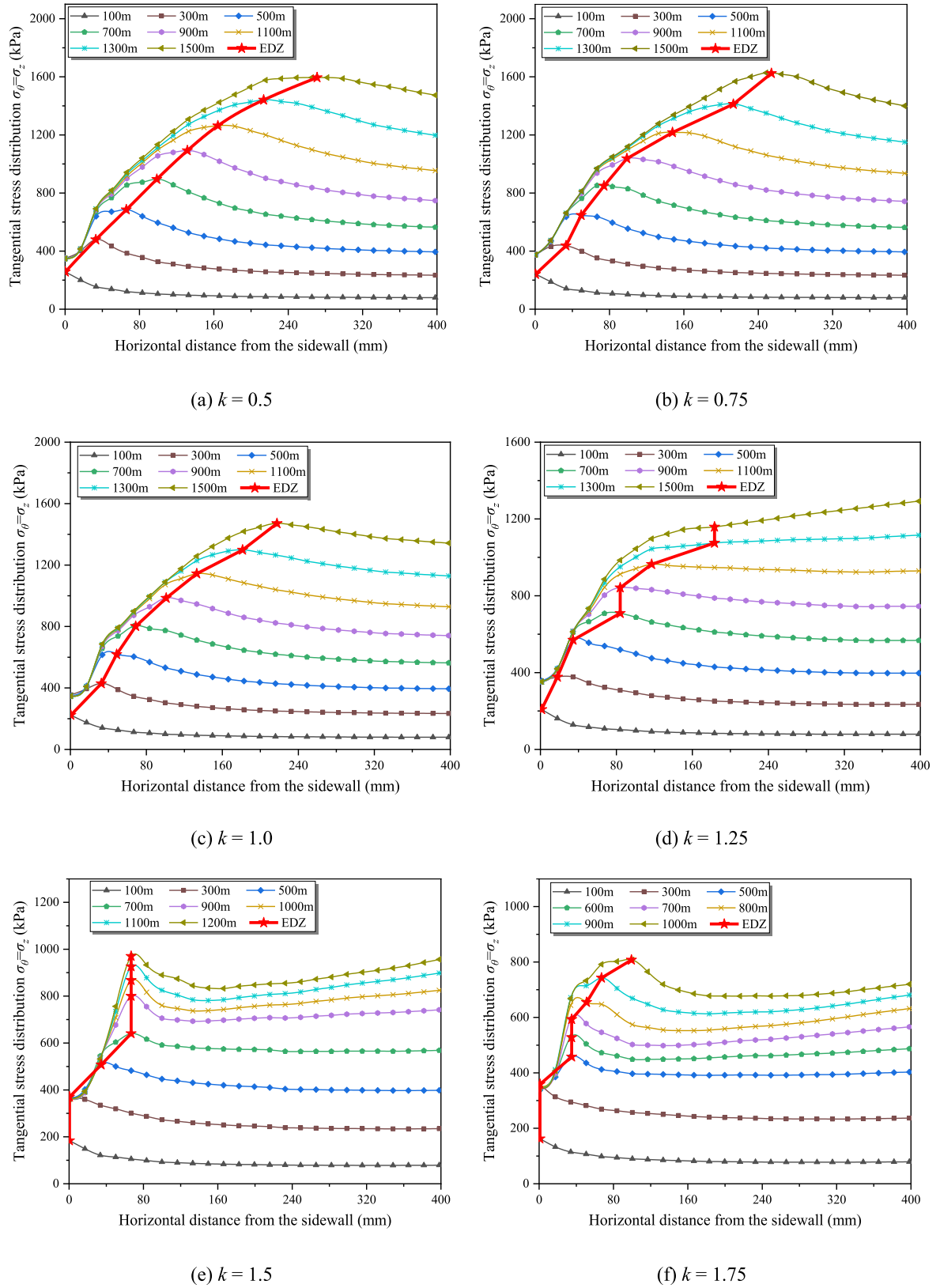
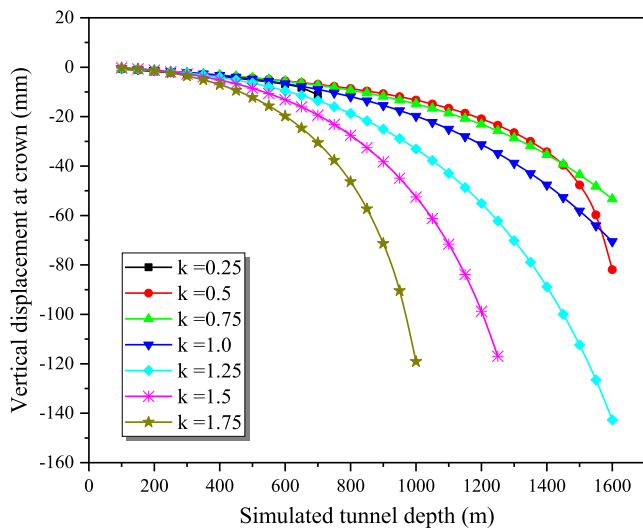


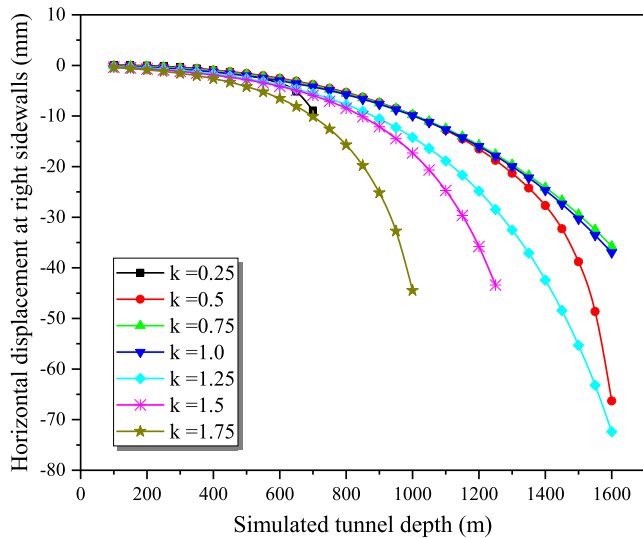
Fig. 12. Distribution of tangential stress horizontally from sidewalls for various values of k .

principal stress axis are the major differences between shallow and deep tunnels, which are also the core factors affecting the deformation and stability of the deep rock tunnels.

(4) The deformation of the surrounding rock has significant nonlinear characteristics related to an increase in buried depth for the unsupported rock tunnels, which can be described by exponential or hyperbolic catenary functions. The nonlinearity is



(a) Vertical deformation of crown settlement



(b) Horizontal deformation of right sidewall

Fig. 13. Deformation of model surrounding tunnels under various buried depths.

Table 10

Fitting accuracy of different functions for the relations of crown settlement and sidewall horizontal deformation with buried depths.

k		Fitting accuracy (R^2)			
		$y = ax + b$	$y = ax^2 + b$	$y = ae^{bx}$	$y = a\cosh(bx)$
0.25	Crown	0.919	0.974	0.986	0.982
	Sidewall	0.735	0.867	0.991	0.991
0.5	Crown	0.721	0.860	0.978	0.978
	Sidewall	0.726	0.870	0.986	0.986
0.75	Crown	0.873	0.977	0.999	0.996
	Sidewall	0.877	0.982	0.994	0.991
1.0	Crown	0.880	0.982	0.996	0.993
	Sidewall	0.873	0.979	0.996	0.993
1.25	Crown	0.842	0.963	0.998	0.997
	Sidewall	0.806	0.941	0.999	0.999
1.5	Crown	0.835	0.956	0.998	0.998
	Sidewall	0.787	0.724	0.999	0.999
1.75	Crown	0.797	0.926	0.998	0.998
	Sidewall	0.757	0.896	0.998	0.998

directly related to the stress conditions, and it increases as the horizontal tectonic stress increases. The deformation characteristics of a shallow tunnel can be described using an approximately linear law, which is not applicable for deep rock tunnels. Thus, the design theories and practical experiences collected from shallow tunnels dominated by gravity stress are not applicable or suitable for deep tunnels.

Funding

This work was supported by the Special Fund for Basic Research on Scientific Instruments of the National Natural Science Foundation of China [41827807]; Emeishan-Hanyuan Highway Program [SRIG2019GG0004]; China Railway First Survey and Design Institute Group Co., Ltd [Grant No. 19–21–1].

CRediT authorship contribution statement

Wuqiang Cai: Investigation, Methodology, Software, Data curation, Formal analysis, Writing – original draft. **Hehua Zhu:** Supervision, Conceptualization, Project administration, Resources. **Wenhao Liang:** Supervision, Conceptualization. **BaThao Vu:** Methodology. **Wei Wu:** Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Al-Ajmi, A.M., Zimmerman, R.W., 2005. Relation between the Mogi and the Coulomb failure criteria. *Int. J. Rock Mech. Min. Sci.* 42 (3), 431–439. <https://doi.org/10.1016/j.ijrmms.2004.11.004>.
- Alejo, L.R., Arzu, J., Bozorgzadeh, N., Harrison, J.P., 2017. Triaxial strength and deformability of intact and increasingly jointed granite samples. *Int. J. Rock Mech. Min. Sci.* 95, 87–103. <https://doi.org/10.1016/j.ijrmms.2017.03.009>.
- Cai, W.Q., Zhu, H.H., Liang, W.H., Zhang, L.Y., Wu, W., 2021. A New Version of the Generalized Zhang-Zhu Strength Criterion and a Discussion on Its Smoothness and Convexity. *Rock Mech Rock Eng.* 54 (8), 4265–4281. <https://doi.org/10.1007/s00603-021-02505-z>.
- Cai, W.Q., Zhu, H.H., Liang, W.H., 2022a. Three-dimensional stress rotation and control mechanism of deep tunneling incorporating generalized Zhang-Zhu strength-based forward analysis. *Eng. Geol.* 308, 106806 <https://doi.org/10.1016/j.enggeo.2022.106806>.
- Cai, W.Q., Zhu, H.H., Liang, W.H., 2022b. Three-Dimensional Tunnel Face Extrusion and Reinforcement Effects of Underground Excavations in Deep Rock Masses. *Int. J. Rock Mech. Min. Sci.* 150, 104999 <https://doi.org/10.1016/j.ijrmms.2021.104999>.
- Cai, W.Q., Zhu, H.H., Liang, W.H., Vu, B.T., Su, C.L., Zhang, K.S., Wang, X.J., 2022c. Three-dimensional forward analysis and real-time design of deep tunneling based on digital in-situ testing. *Int. J. Mech. Sci.* 226, 107385 <https://doi.org/10.1016/j.ijmecsci.2022.107385>.
- Cai, W.Q., Zhu, H.H., Liang, W.H., Wang, X.J., Su, C.L., Wei, X.Y., 2023. A post-peak dilatancy model for soft rock and its application in deep tunnel excavation. *J. Rock Mech Geotech.* <https://doi.org/10.1016/j.jrmge.2022.05.014>.
- Chen, Z.Q., He, C., Xu, G.W., Ma, G.Y., Wu, D., 2019. A Case Study on the Asymmetric Deformation Characteristics and Mechanical Behavior of Deep-Buried Tunnel in Phyllite. *Rock Mech Rock Eng.* 52 (11), 4527–4545. <https://doi.org/10.1007/s00603-019-01836-2>.
- Chen, S.D., Tang, D.Z., Tao, S., Xu, H., Li, S., Zhao, J.L., Ren, P.F., Fu, H.J., 2017. In-situ stress measurements and stress distribution characteristics of coal reservoirs in major coalfields in China: Implication for coalbed methane (CBM) development. *Int. J. Coal Geol.* 182, 66–84. <https://doi.org/10.1016/j.coal.2017.09.009>.
- Feng, G.L., Feng, X.T., Chen, B.R., Xiao, Y.X., Zhao, Z.N., 2019. Effects of structural planes on the microseismicity associated with rockburst development processes in deep tunnels of the Jinping-II Hydropower Station, China. *Tunn. Undergr. Sp. Tech.* 84, 273–280. <https://doi.org/10.1016/j.tust.2018.11.008>.
- Gong, F.Q., Si, X.F., Li, X.B., Wang, S.Y., 2019. Experimental Investigation of Strain Rockburst in Circular Caverns Under Deep Three-Dimensional High-Stress

- Conditions. *Rock Mech Rock Eng.* 52 (5), 1459–1474. <https://doi.org/10.1007/s00603-018-1660-5>.
- Guo, S.F., Qi, S.W., Zhan, Z.F., Ma, L.N., Gure, E.G., Zhang, S.S., 2020. Numerical study on the progressive failure of heterogeneous geomaterials under varied confining stresses. *Eng Geol.* 269 (7), 105556 <https://doi.org/10.1016/j.enggeo.2020.105556>.
- Haimson, B.C., 1978. The hydrofracturing stress measuring method and recent field results. *Int. J. Rock Mech. Min. Sci. Geomech. Abstracts.* 15 (4), 167–178. [https://doi.org/10.1016/0148-9062\(78\)91223-8](https://doi.org/10.1016/0148-9062(78)91223-8).
- He, M.C., Qian, Q.H., 2010. *The Basis of Deep Rock Mechanics*. Science press, Beijing.
- He, M.C., Xie, H.P., Peng, S.P., Jiang, Y.D., 2005. Study on rock mechanics in deep mining engineering. *Chin. J. Rock Mech. Eng.* 24 (16), 2803–2813.
- Hoek, E., Brown, E.T., 2019. The Hoek-Brown failure criterion and GSI - 2018 edition. *J. Rock Mech. Geotech.* 11(3), 445–463. [10.1016/j.jrmge.2018.08.001](https://doi.org/10.1016/j.jrmge.2018.08.001).
- Hoek, E., Brown, E.T., 1997. Practical estimates of rock mass strength. *Int. J. Rock Mech. Min. Sci.* 34 (8), 1165–1186. [https://doi.org/10.1016/S1365-1609\(97\)80069-X](https://doi.org/10.1016/S1365-1609(97)80069-X).
- Hoek, E., Carranza-Torres, C., Corkum, B., 2002. Hoek-Brown failure criterion-2002 edition. *Proc. of NARMS-Tac.* 1 (1), 267–273.
- Hudson, J., 1992. *Rock Engineering Systems: Theory and Practice*. Ellis Horwood, New York.
- Jiang, H., Xie, Y.L., 2012. A new three-dimensional Hoek-Brown strength criterion. *Acta Mech Sinica-Proc.* 28 (2), 393–406. <https://doi.org/10.1007/s10409-012-0054-2>.
- Kang, H., Zhang, X., Si, L., Wu, Y., Gao, F., 2010. In-situ stress measurements and stress distribution characteristics in underground coal mines in China. *Eng Geol.* 116 (3–4), 333–345. <https://doi.org/10.1016/j.enggeo.2010.09.015>.
- Li, G., Ma, F.S., Guo, J., Zhao, H.J., Liu, G., 2020. Study on deformation failure mechanism and support technology of deep soft rock roadway. *Eng Geol.* 264, 105262 <https://doi.org/10.1016/j.enggeo.2019.105262>.
- Li, Y.J., Zhang, D.L., Fang, Q., Yu, Q.C., Xia, L., 2014. A physical and numerical investigation of the failure mechanism of weak rocks surrounding tunnels. *Comput. Geotech.* 61, 292–307. <https://doi.org/10.1016/j.compgeo.2014.05.017>.
- Ma, X.D., Haimson, B.C., 2016. Failure characteristics of two porous sandstones subjected to true triaxial stresses. *J. Geophys Res-Sol Ea.* 121 (9), 6477–6498. <https://doi.org/10.1002/2016jb012979>.
- Ma, T.H., Tang, C.A., Tang, L.X., Zhang, W.D., Wang, L., 2015. Rockburst characteristics and microseismic monitoring of deep-buried tunnels for Jinping II Hydropower Station. *Tunn Undergr Sp Tech.* 49, 345–368. <https://doi.org/10.1016/j.tust.2015.04.016>.
- Meng, L.B., Li, T.B., Jiang, Y., Wang, R., Li, Y.R., 2013. Characteristics and mechanisms of large deformation in the Zhegu mountain tunnel on the Sichuan-Tibet highway. *Tunn Undergr Sp Tech.* 37, 157–164. <https://doi.org/10.1016/j.tust.2013.03.009>.
- Mogi, K., 1967. Effect of the intermediate principal stress on rock failure. *J. Geophys. Res.* 72 (20), 5117–5131. <https://doi.org/10.1029/JZ072i020p05117>.
- Mogi, K., 1971. Fracture and flow of rocks under high triaxial compression. *J. Geophys. Res.* 76 (5), 1255–1269.
- Mogi, K., 2006. *Experimental Rock Mechanics*. CRC Press, London.
- Pan, X., Hudson, J., 1988. A simplified three dimensional Hoek-Brown yield criterion, ISRM International Symposium. International Society for Rock Mechanics and Rock Engineering.
- Pan, P.Z., Feng, X.T., Hudson, J.A., 2012. The influence of the intermediate principal stress on rock failure behaviour: A numerical study. *Eng Geol.* 124, 109–118. <https://doi.org/10.1016/j.enggeo.2011.10.008>.
- Priest, S.D., 2005. Determination of shear strength and three-dimensional yield strength for the hoek-brown criterion. *Rock Mech. Rock Eng.* 38 (4), 299–327. <https://doi.org/10.1007/s00603-005-0056-5>.
- Qi, S.W., Wu, F.Q., Yan, F.Z., Lan, H.X., 2004. Mechanism of deep cracks in the left bank slope of Jinping first stage hydropower station. *Eng Geol.* 73 (1–2), 129–144. <https://doi.org/10.1016/j.enggeo.2003.12.005>.
- Qian, Q.H., Zhou, X.P., 2018. Failure Behaviors and Rock Deformation During Excavation of Underground Cavern Group for Jinping I Hydropower Station. *Rock Mech Rock Eng.* 51 (8), 2639–2651. <https://doi.org/10.1007/s00603-018-1518-x>.
- Singh, J., Ramamurthy, T., Rao, G.V., 1989. Strength of rocks at depth, International Symposium on Rock at Great Depth. Publ Rotterdam: A A Balkema, Publ Rotterdam: A A Balkema 37–44.
- Vu, B.T., Zhu, H., Xu, Q., 2016. Study on failure processes of rock tunnel by model tests and numerical analyses, Geotechnics for Sustainable Infrastructure Development - Geotec Hanoi 2016.
- Vu, B.T., 2014. Investigation on progressive failure of deep weak rock tunnels by physical model tests and numerical analyses, Department of Geotechnical Engineering, College of Civil Engineering. Tongji University, Shanghai.
- Wagner, H., 2019. Deep Mining: A Rock Engineering Challenge. *Rock Mech Rock Eng.* 52 (5), 1417–1446. <https://doi.org/10.1007/s00603-019-01799-4>.
- Wang, S.S., Xu, W.Y., Yan, L., Feng, X.T., Xie, W.C., Chen, H.J., 2020. Experimental investigation and failure mechanism analysis for dacite under true triaxial unloading conditions. *Eng Geol.* 264, 105407 <https://doi.org/10.1016/j.enggeo.2019.105407>.
- Wang, B., Zhu, J.B., Wu, A.Q., Liu, X.H., 2010. Experimental validation of nonlinear strength property of rock under high geostress. *Chin. J. Rock Mech. Eng.* 29 (3), 542–548.
- Wu, K., Shao, Z.S., Jiang, Y.L., Zhao, N.N., Qin, S., Chu, Z.F., 2023. Determination of stiffness of circumferential yielding lining considering the shotcrete hardening property. *Rock Mech Rock Eng.* 56, 1–12. <https://doi.org/10.1007/s00603-022-03122-0>.
- Xie, H.P., Gao, M.Z., Zhang, R., Peng, G.Y., Wang, W.Y., Li, A.Q., 2019. Study on the Mechanical Properties and Mechanical Response of Coal Mining at 1000m or Deeper. *Rock Mech. Rock Eng.* 52 (5), 1475–1490. <https://doi.org/10.1007/s00603-018-1509-y>.
- Yan, P., Lu, W.B., Chen, M., Hu, Y.G., Zhou, C.B., Wu, X.X., 2015. Contributions of In-Situ Stress Transient Redistribution to Blasting Excavation Damage Zone of Deep Tunnels. *Rock Mech Rock Eng.* 48 (2), 715–726. <https://doi.org/10.1007/s00603-014-0571-3>.
- Zhang, L., 2008. A generalized three-dimensional Hoek-Brown strength criterion. *Rock Mech Rock Eng.* 41 (6), 893–915. <https://doi.org/10.1007/s00603-008-0169-8>.
- Zhang, L.Y., Zhu, H.H., 2007. Three-dimensional Hoek-Brown strength criterion for rocks. *J. Geotech Geoenviron.* 133 (9), 1128–1135. [https://doi.org/10.1061/\(ASCE\)1090-0241\(2007\)133:9\(1128\)](https://doi.org/10.1061/(ASCE)1090-0241(2007)133:9(1128)).
- Zhu, H.H., Yan, J.X., Liang, W.H., 2019. Challenges and Development Prospects of Ultra-Long and Ultra-Deep Mountain Tunnels. *Engineering-Proc.* 5 (3), 384–392. <https://doi.org/10.1016/j.eng.2019.04.009>.
- Zhu, H.H., Cai, W.Q., Liang, W.H., 2023. GZZ Strength-based Three-dimensional Analysis Theory and Stress-controlled Design Method in Deep Tunneling. *Chin. J. Rock Mech. Eng.* 42 (1), 1–27.